

ON THE PITFALLS OF VERBALIZED CONFIDENCE PRIORS FOR CALIBRATING REASONING LLMs

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ABSTRACT

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1 INTRODUCTION

Reasoning large language models (LLMs) increasingly solve complex problems by generating a reasoning trajectory followed by the answer (Yang et al., 2025). Despite these advances, the answer accuracy alone does not ensure reliable uncertainty estimates: a model may produce plausible reasoning trajectory yet remain overconfident in an incorrect answer (Geng et al., 2024; Liu et al., 2025). In principle, confidence calibration requires the reported confidence to reflect the empirical probability that the answer is correct, which in turn enables applications such as hallucination detection (Wu et al., 2026) and adaptive test-time scaling (Yang et al., 2026).

Verbalized confidence obtains such an answer-level score by asking the model to emit a numerical probability with its final answer. Unlike confidence estimators based on token probabilities, hidden states, or repeated sampling (Luo et al., 2026; Srey et al., 2026; Wang & Stengel-Eskin, 2026), it can be produced in a single black-box generation (Xiong et al., 2024). Recent work trains this output with supervised fine-tuning (SFT), using targets derived from correctness or rollout statistics (Lin et al., 2022; Li et al., 2025), and with reinforcement learning (RL), by adding calibration terms to answer rewards (Xu et al., 2024; Damani et al., 2026; Ma et al., 2026; Yang et al., 2026).

2 PRELIMINARIES

2.1 VERBALIZED CONFIDENCE FOR REASONING LLMs

In this work, we study confidence calibration for reasoning language models that verbalize a numerical confidence score (Tian et al., 2023; Xiong et al., 2024). Given a question x , the LLM is tasked with producing a completion $o = (r, a, c)$, where r is a reasoning trace and a is an answer. $c \in [0, 1]$ is a post-answer confidence value and represents the model’s estimated probability that a is correct.

Reinforcement Learning with Verifiable Rewards (RLVR). RLVR formulates the LLM as a policy π_θ parameterized by θ . Given a question x , the policy generates a rollout o , and a task verifier assigns it a binary correctness label $z = V(x, a) \in \{0, 1\}$. Typically, the policy is optimized using Group Relative Policy Optimization (GRPO) (Shao et al., 2024). In practice, GRPO samples G rollouts $\{o_i\}_{i=1}^G$ for each question, and the optimization objective can then be formulated as:

$$J_{\text{GRPO}}(\theta) = \mathbb{E} \left[\frac{1}{G} \sum_{i=1}^G \frac{1}{L_i} \sum_{t=1}^{L_i} \min \left(\rho_{i,t}(\theta) \hat{A}_i, \text{clip}(\rho_{i,t}(\theta), 1 - \epsilon_{\text{clip}}, 1 + \epsilon_{\text{clip}}) \hat{A}_i \right) \right]. \quad (1)$$

Here $o_i = (o_{i,1}, \dots, o_{i,L_i})$, and $\rho_{i,t}(\theta) = \pi_\theta(o_{i,t} \mid x, o_{i,<t}) / \pi_{\theta_{\text{old}}}(o_{i,t} \mid x, o_{i,<t})$ is the policy ratio; ϵ_{clip} is the clipping parameter. Given the reward $R_i = R(o_i)$ and a small constant $\epsilon > 0$ for numerical stability, the group-relative advantage is:

$$\hat{A}_i = \frac{R_i - \bar{R}}{\sigma_R + \epsilon}, \quad \bar{R} = \frac{1}{G} \sum_{j=1}^G R_j, \quad \sigma_R = \text{std}(\{R_j\}_{j=1}^G). \quad (2)$$

Confidence-Aware Rewards. For verbalized confidence training, a confidence-based scoring term is typically added to the reward. For example, RLCR (Damani et al., 2026) adds the negative Brier loss to the binary correctness reward:

$$R_i = z_i - (c_i - z_i)^2, \quad (3)$$

where c_i denotes the reported confidence and $z_i = V(x, a_i) \in \{0, 1\}$ denotes the correctness label of rollout i . This reward favors correct answers while assigning higher scores to confidence values that better match their correctness labels, thereby increasing their probability under GRPO.

2.2 CALIBRATION METRICS

A useful verbalized confidence score should be both calibrated and discriminative (Eusebi et al., 2026). To formalize calibration, let C and Z denote the confidence and correctness random variables, respectively. A perfectly calibrated score satisfies $\Pr(Z = 1 \mid C = p) = p$ for every $p \in [0, 1]$. To quantify the degree of miscalibration, **Expected Calibration Error (ECE)** (Guo et al., 2017) is defined as $\mathbb{E}_C[|\Pr(Z = 1 \mid C) - C|]$. In practice, after partitioning N examples into B confidence bins $\{\mathcal{I}_b\}_{b=1}^B$, we estimate ECE as

$$\text{ECE} = \sum_{b=1}^B \frac{|\mathcal{I}_b|}{N} |\text{acc}(\mathcal{I}_b) - \text{conf}(\mathcal{I}_b)|. \quad (4)$$

Here, $\text{acc}(\mathcal{I}_b)$ and $\text{conf}(\mathcal{I}_b)$ are the mean correctness and confidence in bin b , respectively. While ECE measures calibration at the bin level, the **Brier score** (Brier, 1950) measures the squared error between confidence and correctness for each instance,

$$\text{Brier} = \frac{1}{N} \sum_{j=1}^N (c_j - z_j)^2, \quad (5)$$

where lower values indicate better probabilistic predictions for both metrics. These metrics assess calibration, but neither directly evaluates discrimination as a ranking property. For example, assigning every response a constant confidence equal to the overall accuracy yields zero ECE while giving correct and incorrect responses identical scores (Eusebi et al., 2026). We therefore measure the discrimination using **AUROC**, which evaluates whether correct responses receive higher confidence than incorrect ones.

3 MOTIVATION

Confidence-aware RL adds a calibration reward that scores each response’s confidence against its correctness (Xu et al., 2024; Damani et al., 2026; Ma et al., 2026; Yang et al., 2026). Whether this reward alone produces well-calibrated confidence depends on the confidence values the policy actually samples. We therefore examine this distribution before RL, during training, and at test time.

Confidence priors concentrate on a few high values. We refer to the confidence distribution before RL as the model’s *confidence prior*. To characterize this prior, we sample 10 rollouts per question from Qwen3-8B (Yang et al., 2025), Gemma4-E2B-Instruct (Abd et al., 2026), and OLMo3-7B-Instruct (Ettinger et al., 2025) on DeepScaleR-Train. Figure 1 presents their confidence distributions for correct and incorrect responses. All three models concentrate their confidence on a few high values, with fewer than 5% of outputs below 0.5. These base models also assign high confidence even to incorrect responses, suggesting that their verbalized confidence does not reliably reflect answer correctness. RL therefore starts from a concentrated confidence prior that rarely produces low-confidence samples.

RL improves calibration but confidence concentration persists. To examine whether confidence-aware RL overcomes this concentration, we train Qwen3-8B with RLCR (Damani et al., 2026). As shown in Figure 2(a), RLCR reduces Brier loss over training, showing that the calibration reward can be optimized. However, Figure 2(b) shows that the model continues to sample a small set of confidence values for each question, with correspondingly low confidence token entropy throughout training (see Appendix B for training curves across RL objectives).

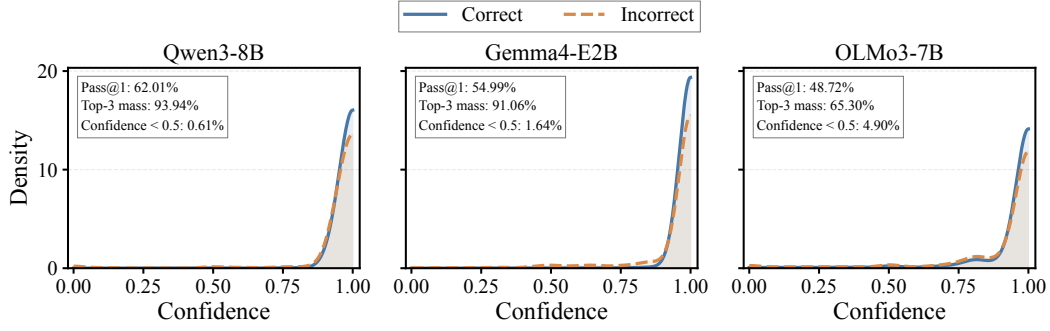


Figure 1: Confidence distributions for correct and incorrect responses from three base models on DeepScaleR-Train. Each distribution is normalized separately; annotations report overall Pass@1, Top-3 mass, and the proportion of confidence outputs below 0.5, pooling both groups. High confidence persists even when answers are incorrect.

To better understand why this concentration persists during training, we formulate confidence generation as action selection under a softmax policy (Garg et al., 2022). Since the LLM decodes each confidence value autoregressively, the induced distribution over the finite set of confidence values is itself a softmax distribution.

Proposition 1 (Policy gradients under confidence concentration). *Given a question x and a finite set of confidence actions $\{c_1, \dots, c_K\} \subseteq [0, 1]$, let $\ell = (\ell_1, \dots, \ell_K)$ denote the policy’s logits for these actions and $\pi_i = \Pr(C = c_i | x) = \exp(\ell_i) / \sum_{j=1}^K \exp(\ell_j)$. For $Z \in \{0, 1\}$ indicating the correctness of a rollout sampled for x , define the fixed expected reward $r_i = -\mathbb{E}[(c_i - Z)^2 | x]$. Let $J_x(\ell) = \sum_i \pi_i r_i$ and $\Delta_x = \max_i r_i - \min_i r_i$. Then*

$$\frac{\partial J_x}{\partial \ell_i} = \pi_i (r_i - J_x), \quad \left| \frac{\partial J_x}{\partial \ell_i} \right| \leq \pi_i \Delta_x. \quad (6)$$

If $\pi_k = 1 - \varepsilon$ for some $k \in \{1, \dots, K\}$, then

$$\|\nabla_{\ell} J_x\|_1 \leq 2\varepsilon \Delta_x. \quad (7)$$

Proposition 1 shows that a rarely sampled confidence value has a small expected logit gradient even when choosing it would improve the expected reward. As the confidence distribution approaches a single choice, the entire logit gradient vanishes, including when that choice is suboptimal. A concentrated confidence distribution therefore impedes exploration of better-calibrated outputs, consistent with the persistently low distinct-value count and confidence token entropy in Figure 2(b). We provide the proof in Appendix C.1.

Persistent confidence concentration hinders downstream calibration. To examine the implications of persistent confidence concentration for downstream calibration, we evaluate the final RLCR checkpoint on two datasets of contrasting difficulty: Math500 and MinervaMath. As shown in Figure 3, the post-RL model underestimates its accuracy on Math500 (90.5% Pass@1 vs. 84.5% mean confidence) but is markedly overconfident on MinervaMath (36.7% Pass@1 vs. 74.8% mean confidence). Despite a 53.8-point gap in Pass@1 between the two tasks, mean confidence differs by only 9.7 points. The three most frequent values still account for 82.25% and 69.49% of outputs on Math500 and MinervaMath, respectively. The same small set of confidence values therefore persists across tasks of different difficulty, so confidence no longer tracks the model’s accuracy on each task.

To characterize the cost of this miscalibration, we examine the expected Brier loss, which is minimized when confidence equals the conditional probability of correctness (Gneiting & Raftery, 2007; Damani et al., 2026).

Proposition 2 (Brier risk under limited confidence support). *Let $Z \in \{0, 1\}$ indicate answer correctness and let $C \in [0, 1]$ denote the predicted confidence. Given a question x , define $p_x = \Pr(Z = 1 | x)$. Assume that C and Z are conditionally independent given x , consistent with the*

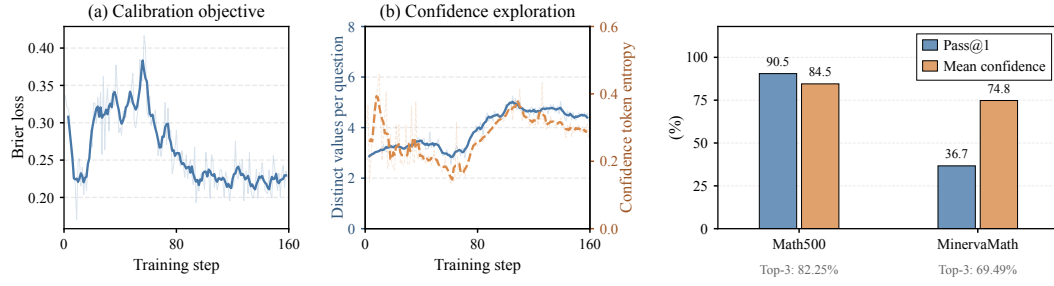


Figure 2: Training dynamics of Qwen3-8B with RLCR. (a) Brier loss. (b) Distinct confidence values per question (eight rollouts; blue, left axis) and confidence token entropy (orange, right axis). Bold curves show centered five-step moving averages; faint curves show per-step measurements. Figure 3: Test-time calibration at step 160 on Math500 and MinervaMath (ID), using four rollouts per question. Bars show Pass@1 and mean confidence; task labels report Top-3 mass.

limited discrimination between correct and incorrect responses observed in Figure 1. For any $\delta > 0$, let $m_\delta(x) = \Pr(|C - p_x| < \delta \mid x)$. Then

$$\mathbb{E}[(C - Z)^2 \mid x] \geq p_x(1 - p_x) + \delta^2(1 - m_\delta(x)). \quad (8)$$

Proposition 2 shows that the lower bound on Brier loss increases as the probability that confidence falls near p_x decreases, so concentration hurts calibration whenever the values it retains sit away from the probability of correctness. In Appendix C.2, we provide the detailed proof, extend the bound to a restricted confidence support, and show that limited support similarly raises the expected error of the closest of G sampled confidence values.

4 METHOD

Section 3 showed that confidence concentration persists during RL, since rarely sampled confidence values have small logit gradients, and that the resulting narrow support hinders downstream calibration. Our key idea is therefore to set the confidence prior by supervision before RL, rather than leaving it to exploration. We propose CalibSFT, a supervised fine-tuning stage for confidence initialization. For each question, CalibSFT samples multiple responses from the base model and assigns each a confidence target that combines the question’s success rate with the response’s own correctness. It supervises confidence on both correct and incorrect responses, but reasoning and answers only on correct ones.

4.1 PROPER-SCORING CONFIDENCE TARGETS

Under a proper scoring rule, the optimal confidence equals the probability that the answer is correct (Gneiting & Raftery, 2007). Since this probability is not observed directly, we estimate it from the base model’s own responses. For each training question x , we sample n responses with correctness labels $z_i \in \{0, 1\}$ and use their success rate as the estimate. However, correct and incorrect responses to the question then share the same confidence target. To support discrimination, we combine the success rate with the response-level label z_i and construct the confidence target of response i as

$$q_x = \frac{1}{n} \sum_{i=1}^n z_i, \quad c_i^* = \lambda q_x + (1 - \lambda) z_i, \quad (9)$$

where the mixing weight $\lambda \in [0, 1]$ balances the two signals. The mixed target is no longer a correctness label, so we examine whether it still rewards the same confidence under a strictly proper scoring rule. Let $p_x = \Pr(Z = 1 \mid x)$ be the success probability of the base model on x , and let Z_i and C_i^* denote the random counterparts of z_i and c_i^* when the n responses are sampled independently. We then have the following proposition.

Proposition 3 (Proper-scoring optimum of the mixed target). *Let $\ell(a, z)$ be a strictly proper scoring loss for $z \in \{0, 1\}$, extended to $t \in [0, 1]$ by $\ell(a, t) = t\ell(a, 1) + (1-t)\ell(a, 0)$. Then $\mathbb{E}[C_i^* | x] = p_x$, and for every a at which ℓ is finite,*

$$\mathbb{E}[\ell(a, C_i^*) | x] = \mathbb{E}[\ell(a, Z_i) | x], \quad \arg \min_{a \in [0, 1]} \mathbb{E}[\ell(a, C_i^*) | x] = p_x. \quad (10)$$

Proposition 3 shows that the mixed target is equivalent to the correctness label in expected loss. In practice, we replace each response’s confidence with c_i^* , preserving its reasoning and answer (see Appendix D.1 for details). The number of responses n controls the variance of the success-rate estimate and the weight λ controls the target’s discrimination within a question; we set $n = 50$ and $\lambda = 0.5$ by default (see Section 5.3 for the λ ablation). We prove the proposition in detail and derive the variance of C_i^* in Appendix C.3.

4.2 CORRECTNESS-CONDITIONAL SUPERVISION

The constructed data contain both correct and incorrect responses. The incorrect ones are kept because their lower targets teach the model to express low confidence in its own errors, whereas imitating their reasoning would fit the model to incorrect solutions. CalibSFT therefore supervises confidence on all responses and restricts reasoning and answer supervision to correct ones.

Formally, let $\tilde{o}$ denote a response whose confidence has been replaced by its constructed target, and let $\mathcal{T}_{\text{ans}}$ and $\mathcal{T}_{\text{conf}}$ denote the positions of its reasoning and answer tokens and its confidence tokens, respectively. The CalibSFT objective is then

$$\begin{aligned} \mathcal{L}_{\text{CalibSFT}} = & (1 - \beta) \mathbb{E}_{\text{correct}} \left[-\frac{1}{|\mathcal{T}_{\text{ans}}|} \sum_{t \in \mathcal{T}_{\text{ans}}} \log \pi_{\theta}(\tilde{o}_t | x, \tilde{o}_{<t}) \right] \\ & + \beta \mathbb{E}_{\text{all}} \left[-\frac{1}{|\mathcal{T}_{\text{conf}}|} \sum_{t \in \mathcal{T}_{\text{conf}}} \log \pi_{\theta}(\tilde{o}_t | x, \tilde{o}_{<t}) \right]. \end{aligned} \quad (11)$$

Here $\mathbb{E}_{\text{correct}}$ averages over the correct responses and $\mathbb{E}_{\text{all}}$ over all of them. Averaging within each segment prevents the longer reasoning sequences from dominating the loss by token count, and β weights confidence supervision against reasoning supervision. We set $\beta = 0.5$, weighting the two terms equally. Since the target values are unevenly distributed across questions, training on the raw data would again concentrate supervision on a few target values. We therefore apply target-balanced sampling, which draws the same number of responses for each target value (Appendix D.1). As an initialization stage, CalibSFT can be combined with different confidence-aware RL methods.

5 EXPERIMENTS

5.1 EXPERIMENTAL SETUP

Baselines. We compare methods spanning answer-only and confidence-aware training. **Base** denotes the checkpoint without task-specific training, while **RLVR** optimizes only answer correctness. We use confidence-aware RL baselines including **RLCR** (Damani et al., 2026), **ReDoubt** (Bani-Harouni et al., 2026), **CoCA** (Li et al., 2026), **DCPO** (Ma et al., 2026), and **C3RL** (Yang et al., 2026). To measure the effect of **CalibSFT**, we evaluate each confidence-aware RL method from both the Base and CalibSFT checkpoints. All methods use Qwen3-8B (Yang et al., 2025) as the default model and follow the **DAPO** training recipe (Yu et al., 2025).

Datasets. We use **DeepScaleR-Preview** (Luo et al., 2025) as the mathematical training dataset. We evaluate on a broad range of in-distribution (ID) and out-of-domain (OOD) datasets spanning different levels of difficulty. For ID mathematical reasoning, we use **DeepScaleR-Eval**, **MATH-500** (Hendrycks et al., 2021), **MinervaMath** (Lewkowycz et al., 2022), **OlympiadBench** (He et al., 2024), **GSM8K** (Cobbe et al., 2021), and **AIME 2024–2026** (Dekoninck et al., 2026). For general reasoning, we use **HotpotQA** (Yang et al., 2018), **MuSiQue** (Trivedi et al., 2022), **TriviaQA** (Joshi et al., 2017), **NQOpen** (Lee et al., 2019), **PopQA** (Mallen et al., 2023), **WebQuestions** (Berant et al., 2013), **DROP** (Dua et al., 2019), and **LiveBenchReasoning** (White et al., 2024).

Table 1: Average performance across 8 ID and 8 OOD datasets. Higher is better for Pass@1 and AUROC; lower is better for Brier and ECE. Each entry averages the datasets within the corresponding split. Base, RLVR, and standalone CalibSFT use a matched 6,144-token evaluation budget.

Method	In-distribution				Out-of-distribution			
	Pass@1 $\uparrow$	AUROC $\uparrow$	Brier $\downarrow$	ECE $\downarrow$	Pass@1 $\uparrow$	AUROC $\uparrow$	Brier $\downarrow$	ECE $\downarrow$
Base	57.39	71.47	34.97	36.26	45.36	65.71	45.15	46.94
RLVR	61.60	73.16	33.48	34.08	43.88	64.19	50.29	51.52
CalibSFT	56.20	79.25	18.01	14.82	45.91	61.38	28.25	24.16
RLCR	62.71	82.78	15.60	16.84	43.96	69.38	31.14	31.17
+ CalibSFT	64.70	86.88	13.48	10.84	45.71	71.31	23.24	18.20
ReDoubt	63.08	81.70	15.94	18.85	43.71	69.88	25.07	23.98
+ CalibSFT	64.99	88.33	13.00	16.20	45.55	73.93	22.42	18.89
CoCA	61.11	80.59	17.25	21.75	43.68	66.79	26.67	24.12
+ CalibSFT	62.65	86.31	16.40	20.21	45.55	71.21	22.95	18.89
DCPO	61.98	78.16	16.38	14.68	43.99	68.13	32.85	32.41
+ CalibSFT	62.40	85.16	13.29	12.20	44.82	72.75	21.73	17.31
C3RL	63.22	67.90	27.80	30.45	43.44	62.19	45.54	48.00
+ CalibSFT	63.32	80.88	15.65	15.18	44.60	63.03	23.89	17.38

Evaluation metrics. We evaluate model performance using complementary metrics that capture answer correctness, confidence discrimination, and absolute calibration. We report **Pass@1** for answer correctness, **AUROC** for ranking correct responses above incorrect responses, and **Brier score** (Brier, 1950) and **ECE** (Guo et al., 2017) for probabilistic calibration.

Implementation details. During evaluation, we use temperature 0.6 and sample 32 rollouts per problem for the AIME benchmarks and 4 rollouts per problem for all other benchmarks. All experiments are conducted on a single node with eight NVIDIA A100 GPUs. We report our hyperparameter setting and training details in Appendix D.

5.2 OVERALL RESULTS

CalibSFT improves calibration across diverse confidence-aware RL methods. As shown in Table 1, initializing RL with CalibSFT reduces Brier score and ECE on mathematical reasoning benchmarks for all three completed paired comparisons: RLCR, CoCA, and DCPO. AUROC also improves for all three methods. We provide detailed per-dataset results in Table 6 (Appendix E.1). For example, CalibSFT reduces DCPO’s ECE from 44.59 to 20.25 while increasing its AUROC from 64.19 to 74.73 on MinervaMath. CalibSFT achieves these gains while largely preserving answer accuracy, with an average Pass@1 gain of 1.32 points across these three RL methods. These results support its use as a plug-and-play initialization for better confidence-aware RL.

CalibSFT generalizes to general reasoning tasks beyond the training domain. To evaluate the generalization of CalibSFT beyond its mathematical domain, we assess the resulting models on eight general reasoning benchmarks. Table 1 shows lower average Brier score and ECE for all three completed paired comparisons on these benchmarks. For example, CalibSFT reduces DCPO’s ECE from 20.53 to 8.55 on MuSiQue. We provide detailed per-dataset results in Table 7 (Appendix E.1).

CalibSFT improves confidence-accuracy correlation across tasks of varying difficulty. To assess whether average confidence tracks accuracy across tasks, we compare each dataset’s average confidence with its accuracy. For each RL method with and without CalibSFT initialization, we compute Pearson and Spearman correlations separately over the eight ID and eight OOD datasets. Table 2 shows that CalibSFT improves Spearman correlation for all three completed RL comparisons on mathematical and general reasoning benchmarks. For example, DCPO’s Spearman correlation increases from 0.619 to 0.976 on mathematical tasks and from 0.548 to 0.643 on general reasoning tasks. Pearson correlation also improves for DCPO and for RLCR on ID tasks, while remaining es-

Table 2: Correlation between accuracy and mean verbalized confidence across ID and OOD datasets. Base and +Ours denote RL without and with CalibSFT initialization, respectively; the higher value is bold. Each coefficient uses eight equally weighted datasets within its split.

Dataset	Metric	RLCR		ReDoubt		CoCA		DCPO		C3RL	
		Base	+Ours	Base	+Ours	Base	+Ours	Base	+Ours	Base	+Ours
ID	Pearson r	0.768	0.963	0.829	0.976	0.750	0.887	0.669	0.996	0.635	0.896
	Spearman ρ	0.619	0.952	0.762	0.976	0.643	0.976	0.619	0.976	0.595	0.833
OOD	Pearson r	0.505	0.729	0.667	0.738	0.447	0.605	0.467	0.611	0.681	0.848
	Spearman ρ	0.548	0.619	0.762	0.643	0.571	0.619	0.548	0.643	0.714	0.929

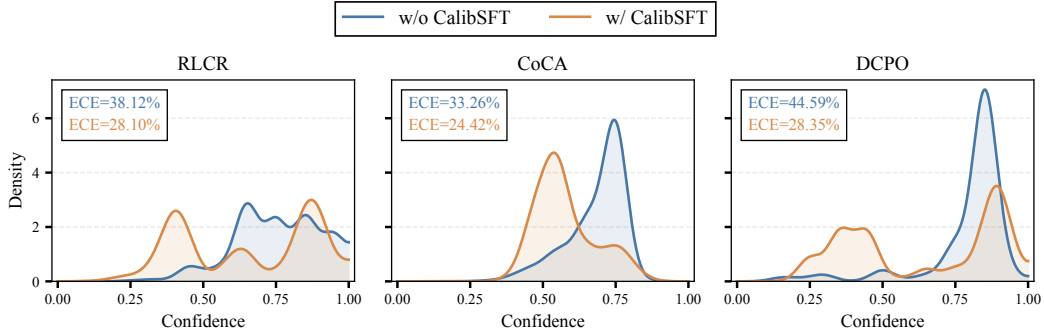


Figure 4: Confidence distributions on MinervaMath for RLCR, CoCA, and DCPO, with four rollouts per question at temperature 0.6. CalibSFT reduces concentration at dominant confidence values and lowers ECE for all three methods.

entially unchanged for RLCR on OOD tasks. These results suggest that CalibSFT helps downstream RL models assign higher average confidence to tasks on which they perform better.

CalibSFT mitigates confidence concentration after downstream RL. Prior work shows that confidence estimates can remain concentrated on a few values even after confidence-aware RL training (Wang et al., 2026; Yang et al., 2026). To examine how CalibSFT initialization influences confidence distribution, we compare RLCR, CoCA, and DCPO on MinervaMath. As shown in Figure 4, models trained without CalibSFT tend to concentrate their confidence on a few values and show higher average confidence than their accuracy. CalibSFT reduces the dominant peaks and brings average confidence closer to accuracy, alleviating both concentration and overconfidence. For example, DCPO’s average confidence decreases from 79.32% to 63.69%, while accuracy increases from 34.93% to 36.58%. ECE decreases for all three methods, from 44.59% to 28.35% for DCPO. We provide additional confidence distribution metrics in Appendix E.7.

The calibration gains of CalibSFT persist across reasoning budgets. To assess whether CalibSFT improves calibration as reasoning progresses rather than the final answer alone, we compare RLCR with and without CalibSFT initialization on AIME 2026 under varying token budgets. As shown in Figure 5, CalibSFT yields substantially lower ECE across all token budgets. It matches the RLCR baseline in accuracy around 5,000 tokens and becomes more accurate at larger budgets. For example, CalibSFT reduces ECE from 22.71 to 10.97 at a 10,000-token budget, while accuracy increases from 47.50% to 52.71%. These results suggest that CalibSFT’s calibration benefits persist as more computation is allocated to reasoning.

5.3 ABLATION STUDIES

Confidence target construction. The construction of confidence targets in CalibSFT is inspired by proper scoring rules, which serve as a foundation in calibration. To verify the effectiveness of this

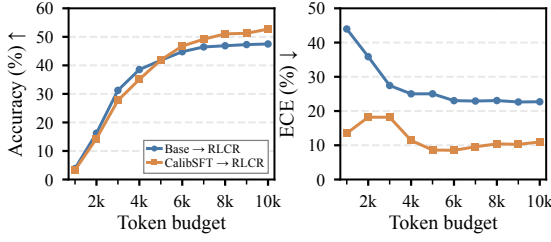


Figure 5: Accuracy (left) and ECE (right) on AIME’26 across token budgets for RLCR with and without CalibSFT initialization. CalibSFT yields substantially lower ECE across all budgets while matching accuracy at medium budgets and improving it at larger budgets.

Target	Pre-SFT targets		Post-SFT predictions	
	AUROC $\uparrow$	Brier $\downarrow$	AUROC $\uparrow$	Brier $\downarrow$
CSFT	50.00	31.67	73.46	19.15
UaIT	52.05	33.17	74.72	24.24
SaySelf	–	41.33	57.69	39.38
LoVeC	75.40	24.71	71.11	32.32
CalibSFT	100.00	7.92	79.75	17.81

Table 3: The calibration comparison between pre-SFT constructed targets and post-SFT predictions. CalibSFT achieves higher AUROC and lower Brier score across ID mathematical benchmarks.

Table 5: Supervision settings and average performance on eight ID benchmarks. A check mark indicates that a segment contributes to the training loss; segment balancing averages the reasoning/answer and confidence losses separately. All variants use the same statically balanced training set and train for 180 steps.

Variant	Reasoning/answer		Confidence		Segment balanced	ID performance			
	Correct	Incorrect	Correct	Incorrect		Pass@1 $\uparrow$	AUROC $\uparrow$	Brier $\downarrow$	ECE $\downarrow$
Confidence-only	–	–	✓	✓	–	43.56	74.62	20.97	17.43
Full-CE	✓	✓	✓	✓	–	65.89	61.66	29.29	27.65
Full-Balanced	✓	✓	✓	✓	✓	65.46	80.67	17.04	13.87
Correct-only	✓	–	✓	–	✓	66.59	81.58	16.95	14.28
CalibSFT	✓	–	✓	✓	✓	65.79	79.30	17.89	14.36

target design, we compare CalibSFT with targets based on group-level correctness (CSFT) (Jang et al., 2025), token probabilities (UaIT) (Liu et al., 2024), semantic consistency (SaySelf) (Xu et al., 2024), and rubric-based LLM judgments (LoVeC) (Zhang et al., 2026), using a shared SFT training set the same optimization objective. We provide details of the subset and target construction in Appendix E.2. We report the results of the target construction in Table 3, which shows the AUROC and Brier score of the constructed targets before SFT and average performance on ID mathematical benchmarks after SFT. CalibSFT yields lower Brier score and higher AUROC than the four alternatives. These results support the effectiveness of our confidence target construction, which accounts for both calibration and discrimination.

Target mixing weight. We construct confidence targets as $c_i^* = \lambda q_x + (1 - \lambda) z_i$, where q_x is the group success rate and z_i is the correctness of an individual response. To evaluate the effect of the mixing weight, we compare $\lambda \in \{0, 0.25, 0.5, 0.75, 1\}$ using the same subset as above and keeping the SFT settings fixed. At $\lambda = 0$, targets are restricted to 0 and 1; this setting yields the highest Brier score and ECE in Table 4. At $\lambda = 1$, correct and incorrect responses to the same question receive identical targets, providing no within-question discrimination signal; post-SFT AUROC is also lower than at $\lambda = 0.5$. Equal weighting achieves the highest AUROC and lowest Brier score and ECE among the tested settings, so we use $\lambda = 0.5$ as the default.

Table 4: Average calibration performance on ID benchmarks across mixing weights. Equal weighting ($\lambda = 0.5$) performs best on all three metrics.

λ	AUROC $\uparrow$	Brier $\downarrow$	ECE $\downarrow$
0	68.10	32.49	32.49
0.25	79.02	18.83	14.21
0.5	79.75	17.81	13.03
0.75	65.93	19.99	13.16
1	73.29	19.90	15.04

Supervision design. [TODO:] Compare confidence-only SFT, full-completion CE, segment-balanced full-completion CE, CalibSFT, and CalibSFT without target-balanced sampling. Hold the training examples, confidence targets, and optimization settings fixed across the supervision variants. Report which completion segments contribute to each objective, followed by their ID/OOD Pass@1 and calibration results. Use this ablation to isolate the effects of retaining answer supervision on correct rollouts, masking incorrect reasoning and answers, supervising confidence on all rollouts, and balancing confidence targets.

6 DISCUSSION

6.1 VERBALIZED CONFIDENCE AND ALTERNATIVE CONFIDENCE SIGNALS

Verbalized confidence provides a directly interpretable uncertainty signal. [TODO:] Compare post-answer verbalized confidence with token- or sequence-level probability, self-consistency, semantic consistency, and a hidden-state probe. Use the same generated answers whenever the signal permits, and report AUROC, Brier score, ECE, inference cost, and model-access requirements. Discuss which gains arise from CalibSFT and which arise from the choice of confidence signal.

6.2 CONFIDENCE FORMATS AND RESOLUTION

CalibSFT can be instantiated with different confidence formats. [TODO:] Compare two-decimal probabilities, discrete confidence levels, and semantic labels such as *low*, *medium*, and *high*. Map all formats to $[0, 1]$ for evaluation and report Pass@1, AUROC, Brier score, and ECE. Include ConfTuner as the representative tokenized-confidence method when its output format and evaluation mapping are matched.

The response count controls the resolution of empirical targets. [TODO:] Fix $\lambda = 0.5$ and compare target construction with $n \in \{10, 50\}$. Report the resulting target coverage, target estimation cost, and final calibration. Use this comparison to separate the effect of a finer confidence grid from the effect of additional responses.

7 DOWNSTREAM APPLICATIONS

7.1 SELECTIVE ANSWERING AND RISK CONTROL

[TODO:] Rank responses by confidence and report risk–coverage curves, AURC, and accuracy at matched coverage. Compare Base, confidence-aware RL, and CalibSFT-initialized RL to test whether the calibration improvements translate into more reliable abstention decisions.

7.2 CONFIDENCE-GUIDED EFFICIENT REASONING

[TODO:] Use confidence to allocate additional generation budget or route uncertain questions to a stronger inference procedure. Compare against fixed-budget and random-routing baselines at matched token cost, reporting accuracy, calibration, and total generated tokens.

8 CONCLUSION

[TODO:] Summarize the concentrated and misaligned confidence prior, the rollout-derived and target-balanced CalibSFT initialization, and its effect on downstream confidence-aware RL. Close with the main ID/OOD evidence and the remaining limitations of rollout-derived confidence targets.

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AUTHOR CONTRIBUTIONS

If you’d like to, you may include a section for author contributions as is done in many journals. This is optional and at the discretion of the authors.

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A RELATED WORK

A.1 CONFIDENCE ESTIMATION IN LLMs

LLM confidence can be estimated from token probabilities, hidden-state probes, agreement across sampled answers, or explicit self-reports (Kadavath et al., 2022; Azaria & Mitchell, 2023; Kuhn et al., 2023; Xiong et al., 2024). Verbalized-confidence methods ask the model to emit a numerical or linguistic confidence score, providing a single-generation estimate without requiring access to internal model representations (Tian et al., 2023; Xiong et al., 2024). Our work studies this post-answer verbalized setting.

A.2 CONFIDENCE-AWARE POST-TRAINING

Supervised approaches train confidence outputs using correctness labels, empirical rollout frequencies, or confidence-token objectives (Lin et al., 2022; Xu et al., 2024; Li et al., 2025). Calibration-aware RL instead incorporates proper scoring rewards or modifies the confidence target and credit assignment. Representative methods include RLCR, Rewarding Doubt, CoCA, DCPO, and CALIBER (Damani et al., 2026; Bani-Harouni et al., 2026; Li et al., 2026; Ma et al., 2026; Finlay et al., 2026). CalibSFT focuses on the supervised initialization supplied to these downstream RL objectives.

B ADDITIONAL ANALYSIS OF THE MOTIVATION

In Section 3, we observe that confidence stays concentrated on a few values while RLCR optimizes its calibration reward. To verify that this behavior is not specific to one objective, we further train Qwen3-8B with CoCA (Li et al., 2026) and DCPO (Ma et al., 2026) under the same setup and track the same quantities.

As shown in Figure 6, we observe the same phenomenon as in Section 3. Between the first and last ten steps, Brier loss falls for RLCR (0.266 to 0.224) and CoCA (0.255 to 0.221) and stays nearly flat for DCPO (0.246 to 0.241), so each objective either improves calibration or holds it steady. Confidence support does not follow. The mean number of distinct confidence values within a group of eight responses never exceeds 5.3 at any step of any run, and confidence token entropy ends below where it started in all three runs. Therefore, none of the three objectives moves the model away from sampling a few confidence values per question, which extends the observation reported for RLCR in Section 3 to all three objectives.

C THEORETICAL ANALYSIS

C.1 PROOF OF PROPOSITION 1

Proof. We fix x and the expected rewards r_i , and differentiate with respect to the confidence logits. The softmax derivative is $\partial\pi_j/\partial\ell_i = \pi_j(\mathbf{1}\{j=i\} - \pi_i)$. It follows that

$$\frac{\partial J_x}{\partial\ell_i} = \sum_j r_j \pi_j (\mathbf{1}\{j=i\} - \pi_i) \quad (12)$$

$$= \pi_i(r_i - J_x). \quad (13)$$

Since J_x is a convex combination of the r_i values, $|r_i - J_x| \leq \Delta_x$, which gives the coordinate-wise bound. If $\pi_k = 1 - \varepsilon$, then

$$|r_k - J_x| = \left| \sum_{i \neq k} \pi_i(r_k - r_i) \right| \leq \varepsilon \Delta_x. \quad (14)$$

Consequently,

$$\|\nabla_{\ell} J_x\|_1 = \pi_k |r_k - J_x| + \sum_{i \neq k} \pi_i |r_i - J_x| \quad (15)$$

$$\leq (1 - \varepsilon)\varepsilon\Delta_x + \varepsilon\Delta_x \quad (16)$$

$$\leq 2\varepsilon\Delta_x. \quad (17)$$

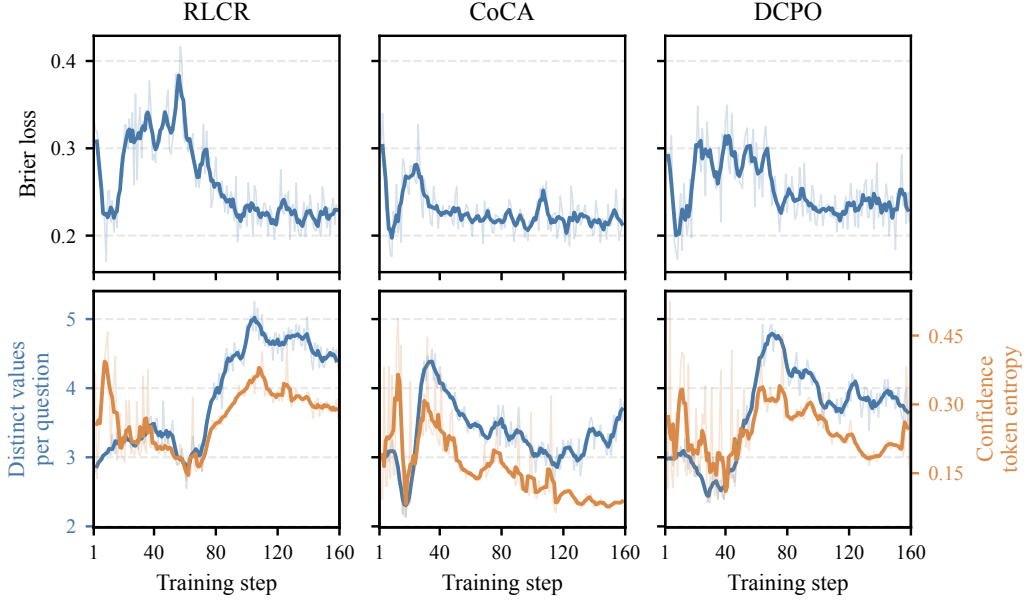


Figure 6: Training dynamics of confidence calibration and support for Base→RLCR, Base→CoCA, and Base→DCPO. The top row shows Brier loss; the bottom row shows distinct confidence values per question (blue, left axis) and confidence token entropy (orange, right axis). Faint curves show per-step measurements and bold curves centered five-step means. Statistics use the groups retained by the corresponding training run, so the sampled questions and responses change across steps.

□

The gradient is the expectation of the likelihood-ratio estimator $R\nabla_{\ell} \log \pi_I$, where $I \sim \pi$ and $R = -(c_I - Z)^2$, with Z drawn from its fixed conditional distribution given x , independently of I . This matches RLCR’s reward $Z - (c_I - Z)^2$ (Preliminaries) up to the term Z , which does not depend on c_I . Assuming $C \perp Z \mid x$ as in Proposition 2, this term contributes an ℓ -independent additive constant to J_x that does not affect this gradient. A baseline depending only on x leaves this expectation unchanged.

C.2 PROOF OF PROPOSITION 2

Proof. Conditional independence gives $\mathbb{E}[Z - p_x \mid x, C] = 0$. Expanding the square therefore yields

$$\mathbb{E}[(C - Z)^2 \mid x] = \mathbb{E}[(Z - p_x)^2 \mid x] + \mathbb{E}[(C - p_x)^2 \mid x] \quad (18)$$

$$= p_x(1 - p_x) + \mathbb{E}[(C - p_x)^2 \mid x]. \quad (19)$$

For any $\delta > 0$, the second term satisfies

$$\mathbb{E}[(C - Z)^2 \mid x] - p_x(1 - p_x) = \mathbb{E}[(C - p_x)^2 \mid x] \quad (20)$$

$$\geq \delta^2 \Pr(|C - p_x| \geq \delta \mid x) \quad (21)$$

$$= \delta^2(1 - m_{\delta}(x)). \quad (22)$$

Thus, assigning little probability mass near p_x incurs excess Brier loss even when confidence values in that neighborhood remain possible. □

Restricted confidence support. As a special case, suppose $\Pr(C \in S_x \mid x) = 1$ for a nonempty set $S_x \subseteq [0, 1]$, and define $d_x = \text{dist}(p_x, S_x) = \inf_{c \in S_x} |c - p_x|$. If $d_x > 0$, then $m_{d_x}(x) = 0$, and Proposition 2 with $\delta = d_x$ gives

$$\mathbb{E}[(C - Z)^2 \mid x] \geq p_x(1 - p_x) + \text{dist}(p_x, S_x)^2. \quad (23)$$

For $d_x = 0$, the same bound follows directly from the decomposition above. If S_x is closed, a predictor that always outputs a nearest point in S_x attains the bound. Confidence values are effectively drawn from a small discrete set throughout this paper (Figure 1), so $\text{dist}(p_x, S_x)$ measures how far the nearest available value falls from p_x , translating the observed concentration directly into excess Brier risk.

Best-of- G sampling under a concentrated policy. Given the policy and a question x , draw G independent confidence samples $C_1, \dots, C_G$ from its conditional distribution. Each sample lies outside $\{c : |c - p_x| < \delta\}$ with probability $1 - m_\delta(x)$, and by independence all G samples do so with probability $(1 - m_\delta(x))^G$. On that event, even the sample closest to p_x has squared error at least δ^2 , so

$$\mathbb{E} \left[\min_{i \leq G} (C_i - p_x)^2 \mid x \right] \geq \delta^2 (1 - m_\delta(x))^G. \quad (24)$$

With the group size $G = 8$ used in our RL runs, a policy that rarely samples near p_x is unlikely to produce a well-calibrated confidence anywhere in the group.

C.3 PROOF OF PROPOSITION 3

We analyze the unrounded targets under the original sampling distribution, before response filtering and target balancing. The random variables Q_x and C_i^* correspond to the observed q_x and c_i^* in Equation 9.

Proof of Proposition 3. Fix x and write $v_x = p_x(1 - p_x)$. Conditional independence gives

$$\mathbb{E}[Q_x \mid x] = \mathbb{E}[Z_i \mid x] = p_x, \quad \text{Var}(Q_x \mid x) = \frac{v_x}{n}. \quad (25)$$

Since Z_i is included in Q_x , their conditional covariance is

$$\text{Cov}(Q_x, Z_i \mid x) = \frac{v_x}{n}. \quad (26)$$

Linearity of expectation gives $\mathbb{E}[C_i^* \mid x] = p_x$. Expanding the variance yields

$$\text{Var}(C_i^* \mid x) = \lambda^2 \frac{v_x}{n} + (1 - \lambda)^2 v_x + 2\lambda(1 - \lambda) \frac{v_x}{n} \quad (27)$$

$$= v_x \left[\frac{1}{n} + (1 - \lambda)^2 \left(1 - \frac{1}{n} \right) \right]. \quad (28)$$

Now fix a at which ℓ is finite. Since $\ell(a, t)$ is affine in t , unbiasedness gives

$$\mathbb{E}[\ell(a, C_i^*) \mid x] = p_x \ell(a, 1) + (1 - p_x) \ell(a, 0) = \mathbb{E}[\ell(a, Z_i) \mid x]. \quad (29)$$

Because ℓ is strictly proper, this expected loss is uniquely minimized at $a = p_x$. Thus the mixture, the group average, and an individual correctness label induce the same expected scoring loss for a prediction based on x . For the Brier loss $\ell(a, z) = (a - z)^2$, the extension is $\ell(a, t) = (a - t)^2 + t(1 - t)$, so the squared loss against a soft target has the same minimizer. The same argument applies to the binary logarithmic loss (Gneiting & Raftery, 2007). $\square$

Effect of the response count and mixing weight. At $\lambda = 0$, the target is the binary label with variance v_x ; at $\lambda = 1$, it is the group average with variance v_x/n . For a fixed $\lambda < 1$, the variance approaches $(1 - \lambda)^2 v_x$ as n increases, reflecting the retained contribution of the individual outcome. Within any observed group, $C_i^* - C_j^* = (1 - \lambda)(Z_i - Z_j)$, so the targets of a correct and an incorrect response differ by $1 - \lambda$.

Conditioning on the generated response. Proposition 3 concerns the success probability p_x before observing a particular response. To make the effect of conditioning explicit, let h_i include x and information from response i , and define $p_{h_i} = \Pr(Z_i = 1 \mid h_i)$. If the other responses are independent of h_i conditional on x , then

$$\mathbb{E}[C_i^* \mid h_i] = \left(1 - \lambda + \frac{\lambda}{n} \right) p_{h_i} + \lambda \left(1 - \frac{1}{n} \right) p_x. \quad (30)$$

The target thus shrinks the response-conditional probability toward the question’s success probability. These results justify the mean and variance of this supervision signal under the original sampling distribution; it does not establish response-conditional calibration after target balancing and token-level SFT. Those effects are evaluated empirically.

D IMPLEMENTATION DETAILS

D.1 TRAINING PIPELINES

All runs use Qwen3-8B as the base model. We randomly hold out 2,000 examples from DeepScaleR-Preview as a test split (DeepScaleR-Eval) and use all remaining examples as the training split (DeepScaleR-Train), which is used to construct the SFT data for CalibSFT and the downstream RL training data.

CalibSFT. To construct CalibSFT targets, we sample 50 responses per question from DeepScaleR-Train with temperature 1.0, top- p 0.95, and a cap of 8,192 tokens. We compute each question’s success rate and combine it with each response’s correctness using $\lambda = 0.5$. Targets are balanced offline when constructing the training split, and we train with the selective completion mask and $\beta = 0.5$: correct responses supervise both the reasoning/answer and confidence segments, whereas incorrect responses supervise only the confidence segment. We train CalibSFT for 180 steps with a learning rate of 2×10^{-6} and 10 warm-up steps.

SFT data construction. We compute each question’s success rate q_x from all sampled responses before filtering. Invalid or low-quality responses are then removed from the SFT data, with q_x kept fixed. For each retained response, we replace the confidence suffix with its target while preserving the sampled reasoning and answer. In the main probability format, the target is rendered to two decimal places in the trailing <confidence> tag. For balanced sampling, we group examples by the numeric target rounded to six decimal places and sample the same number from every nonempty group, using the smallest group size as the default count. We apply this rule either through a per-epoch sampler or when constructing an offline balanced split.

Confidence-aware RL. For downstream RL, each prompt group contains 8 sampled responses, with each response capped at 8,192 tokens. We train with batches of 64 prompt groups for 160 optimization steps. For methods using DAPO dynamic sampling, this corresponds to approximately 600 rollout-generation batches. Dynamic sampling retains only groups with mixed correctness outcomes and discards groups whose responses are all correct or all incorrect. To accelerate training for these methods, we remove questions for which all sampled responses from the base model are correct before RL training begins. Each method retains its own reward and credit assignment.

- **Direct RL baselines.** RLVR, RLCR (Damani et al., 2026), CoCA (Li et al., 2026), and DCPO (Ma et al., 2026) are trained directly from the Qwen3-8B base checkpoint. They share the DAPO training recipe but retain their respective correctness, calibration-target, and credit-assignment objectives.
- **ReDoubt.** ReDoubt is initialized from the trained RLVR checkpoint and then optimized with its clipped log-score confidence objective (Bani-Harouni et al., 2026). When incorporated with CalibSFT, we first train CalibSFT on the RLVR checkpoint and then continue with ReDoubt RL.
- **C3RL.** C3RL (Yang et al., 2026) combines correctness, calibration, and reference-accuracy rewards. C3RL uses the digit confidence format, expressing confidence as an integer from 0 to 9 and mapping it to $[0, 1]$ by dividing by 9. The CalibSFT initialization for C3RL is also trained in this format. We use the base model’s rollout success rates to label each training question as all correct, partially correct, or all incorrect. The reference reward encourages correct answers to previously all-incorrect questions and penalizes incorrect answers to previously all-correct questions. To retain these questions and their reference-reward signals, we disable DAPO dynamic sampling and all-correct filtering, using the full RL training split for C3RL both with and without CalibSFT initialization.

D.2 EVALUATION PROTOCOL

For the main results, we use temperature 0.6 and sample four rollouts per question on non-AIME mathematical and OOD benchmarks and 32 rollouts per question on AIME. Pass@1 is the mean correctness over these sampled rollouts. The supervision and target ablations retain their recorded evaluation settings. Confidence is parsed from the final probability field and normalized to $[0, 1]$. We use 10 equal-width bins for calculating ECE. Unless otherwise noted, RL results use the final checkpoint after 160 optimization steps, while SFT-only results use the checkpoint at the end of their respective SFT runs.

E DETAILED EXPERIMENTAL RESULTS

E.1 PER-DATASET MAIN RESULTS

Table 1 reports split-level averages. The following tables expose the same metrics for every benchmark; the final cell in each split is the unweighted average over its datasets.

E.2 ABLATION OF SFT CONFIDENCE TARGETS

To evaluate the advantage of confidence targets derived from a proper scoring rule, we compare CalibSFT with targets based on group-level correctness (CSFT) (Jang et al., 2025), token probabilities (UaIT) (Liu et al., 2024), semantic consistency (SaySelf) (Xu et al., 2024), and rubric-based LLM judgments (LoVeC) (Zhang et al., 2026). For a fair comparison, we construct the training datasets from a shared response pool. SaySelf uses its correct responses, following its SFT setup. We sample questions uniformly across empirical success rates so that the comparison covers both easier and harder questions.

For each question in DeepScaleR-Train, we use 10 responses generated by Qwen3-8B. With $z_i \in \{0, 1\}$ denoting the response correctness, the empirical success rate is $q_x = \frac{1}{10} \sum_{i=1}^{10} z_i$. We select 300 questions at each $q_x \in \{0.1, 0.2, \dots, 0.9\}$ and retain one correct and one incorrect response per question. Finally, We have 5,400 responses from 2,700 questions, with equal numbers of correct and incorrect responses at each success rate.

On this shared training subset, we construct the five confidence targets as follows:

- **CSFT.** Both responses receive the group success rate, $c_i^* = q_x$.
- **UaIT.** The target is $c_i^* = \exp(-\text{NLL}_i)$, where NLL_i is the mean negative log-likelihood of the sampled completion tokens under the generating model.
- **SaySelf.** SaySelf uses an embedding model to group semantically similar responses and derives confidence from cluster frequency. We embed the ten responses to each question using Qwen3-Embedding-8B and cluster them with a cosine similarity threshold of 0.98. Since SaySelf uses only correct responses for SFT, we retain the 2,700 selected correct responses and assign each the frequency of its cluster, $c_i^* = |\mathcal{M}_i|/10$, where $\mathcal{M}_i$ is the cluster containing response i .
- **LoVeC.** LoVeC uses an LLM judge to assess the factuality of generated statements and supervise their confidence estimates. Our rubric assigns 10 to correct answers with sound reasoning, 8–9 to correct answers with minor reasoning gaps, 5–7 to partially correct or ambiguous responses, 2–4 to meaningful but incorrect attempts, and 0–1 to invalid or clearly wrong responses. We normalize its score $s_i \in \{0, \dots, 10\}$ to obtain $c_i^* = s_i/10$.

Table 8: Training-target distribution (%) over ten equal-width bins in the confidence target ablation.

Interval	CSFT	UaIT	SaySelf	LoVeC	CalibSFT
[0.0, 0.1]	11.11	0.00	22.07	27.33	11.11
(0.1, 0.2]	11.11	0.00	14.41	3.78	11.11
(0.2, 0.3]	11.11	0.00	11.93	8.17	11.11
(0.3, 0.4]	11.11	0.00	10.22	0.20	11.11
(0.4, 0.5]	11.11	0.00	9.04	1.37	5.56
(0.5, 0.6]	11.11	0.00	8.26	0.63	11.11
(0.6, 0.7]	11.11	6.48	8.89	0.22	11.11
(0.7, 0.8]	11.11	56.06	6.63	1.54	11.11
(0.8, 0.9]	11.11	35.15	5.56	15.80	11.11
(0.9, 1.0]	0.00	2.31	3.00	40.96	5.56

Table 6: Per-dataset results on the in-domain mathematical benchmarks. Base, RLVR, and standalone CalibSFT use 6,144 tokens; downstream RL models use 16,384 tokens. Temperature is 0.6, with four rollouts per non-AIME question and 32 per AIME question.

(a) DeepScaleR-Eval					(b) MATH-500					(c) MinervaMath				
Method	P@1	AUROC	Brier	ECE	Method	P@1	AUROC	Brier	ECE	Method	P@1	AUROC	Brier	ECE
Base	60.84	66.31	34.09	34.06	Base	83.90	63.80	14.78	13.87	Base	39.89	63.26	53.79	55.28
RLVR	70.15	68.11	26.94	26.37	RLVR	89.15	67.69	10.13	8.76	RLVR	37.96	60.92	57.18	58.40
CalibSFT	60.88	73.76	23.60	17.57	CalibSFT	83.65	82.11	14.70	11.53	CalibSFT	39.89	57.50	30.33	24.20
RLCR	71.24	75.33	16.65	5.14	RLCR	90.50	76.39	7.54	7.43	RLCR	36.67	66.67	35.91	38.12
+ CalibSFT	72.47	79.34	15.33	4.58	+ CalibSFT	90.15	89.79	6.56	3.69	+ CalibSFT	37.04	76.29	26.81	26.33
ReDoubt	71.53	75.92	17.13	10.28	ReDoubt	90.05	74.85	9.82	15.01	ReDoubt	37.59	66.65	27.92	25.37
+ CalibSFT	72.36	78.06	16.06	6.67	+ CalibSFT	89.45	89.64	9.60	16.74	+ CalibSFT	36.21	83.73	21.06	22.86
CoCA	70.16	73.17	17.73	8.51	CoCA	89.85	72.69	10.64	18.18	CoCA	35.20	65.28	32.43	33.26
+ CalibSFT	71.45	78.71	16.41	7.92	+ CalibSFT	89.85	87.29	9.09	15.49	+ CalibSFT	36.95	78.62	23.47	24.42
DCPO	71.45	71.74	17.30	6.50	DCPO	89.30	70.77	7.90	6.27	DCPO	34.93	64.19	41.78	44.59
+ CalibSFT	70.46	77.71	16.06	5.95	+ CalibSFT	90.20	84.89	7.68	7.70	+ CalibSFT	36.86	74.73	23.33	20.25
C3RL	70.65	62.87	24.64	25.05	C3RL	89.40	60.22	9.42	9.75	C3RL	37.22	56.00	58.42	59.49
+ CalibSFT	71.05	71.91	19.93	14.72	+ CalibSFT	87.70	84.76	10.37	9.36	+ CalibSFT	35.85	77.64	20.14	16.95
(d) OlympiadBench					(e) GSM8K					(f) AIME2024				
Method	P@1	AUROC	Brier	ECE	Method	P@1	AUROC	Brier	ECE	Method	P@1	AUROC	Brier	ECE
Base	54.41	69.82	38.93	40.13	Base	94.35	67.82	5.20	3.39	Base	48.23	81.93	39.61	42.50
RLVR	56.31	72.84	39.35	39.98	RLVR	94.64	67.97	5.06	3.05	RLVR	52.92	82.05	39.49	41.15
CalibSFT	51.67	79.33	22.48	20.62	CalibSFT	94.31	70.24	17.90	20.78	CalibSFT	45.42	89.96	11.93	6.03
RLCR	57.83	81.78	19.04	12.50	RLCR	94.81	78.88	4.77	5.58	RLCR	59.48	93.59	13.17	19.70
+ CalibSFT	58.46	89.71	16.09	13.25	+ CalibSFT	94.67	82.60	4.64	2.72	+ CalibSFT	64.17	92.42	10.97	9.79
ReDoubt	57.57	80.67	18.18	8.36	ReDoubt	94.96	78.40	7.07	16.27	ReDoubt	55.62	91.94	15.40	22.77
+ CalibSFT	58.35	91.01	14.96	16.88	+ CalibSFT	94.69	80.28	7.50	17.62	+ CalibSFT	61.56	93.81	11.89	14.37
CoCA	56.38	80.19	20.14	14.63	CoCA	94.58	69.00	8.94	21.02	CoCA	53.85	92.22	16.27	21.37
+ CalibSFT	56.19	90.47	18.09	20.16	+ CalibSFT	94.56	71.85	8.26	18.10	+ CalibSFT	58.54	95.41	16.34	23.56
DCPO	57.05	78.43	21.85	17.75	DCPO	94.48	68.34	5.50	9.13	DCPO	57.92	88.84	12.32	8.47
+ CalibSFT	57.46	88.53	15.72	12.14	+ CalibSFT	94.45	73.40	5.80	7.48	+ CalibSFT	58.33	93.89	11.02	11.85
C3RL	57.68	62.66	35.85	37.52	C3RL	94.37	57.25	5.33	5.24	C3RL	59.69	80.93	25.25	30.19
+ CalibSFT	56.64	84.53	17.91	18.60	+ CalibSFT	94.29	64.91	5.26	5.08	+ CalibSFT	64.27	85.24	14.04	12.99
(g) AIME2025					(h) AIME2026					(i) Overall (ID)				
Method	P@1	AUROC	Brier	ECE	Method	P@1	AUROC	Brier	ECE	Method	P@1	AUROC	Brier	ECE
Base	35.00	82.75	48.25	52.81	Base	42.50	76.08	45.13	48.07	Base	57.39	71.47	34.97	36.26
RLVR	43.44	84.95	45.87	49.00	RLVR	48.23	80.70	43.82	45.92	RLVR	61.60	73.16	33.48	34.08
CalibSFT	34.27	92.48	9.95	8.02	CalibSFT	39.48	88.65	13.22	9.79	CalibSFT	56.20	79.25	18.01	14.82
RLCR	43.33	95.53	13.60	23.06	RLCR	47.81	94.10	14.08	23.19	RLCR	62.71	82.78	15.60	16.84
+ CalibSFT	48.02	92.40	13.87	14.28	+ CalibSFT	52.60	92.48	13.57	12.04	+ CalibSFT	64.70	86.88	13.48	10.84
ReDoubt	44.90	94.22	15.19	26.51	ReDoubt	52.40	90.93	16.78	26.20	ReDoubt	63.08	81.70	15.94	18.85
+ CalibSFT	51.77	96.32	11.28	16.79	+ CalibSFT	55.52	93.82	11.66	17.69	+ CalibSFT	64.99	88.33	13.00	16.20
CoCA	40.31	95.39	17.04	29.31	CoCA	48.54	96.74	14.84	27.72	CoCA	61.11	80.59	17.25	21.75
+ CalibSFT	45.00	94.21	19.99	26.43	+ CalibSFT	48.65	93.96	19.52	25.63	+ CalibSFT	62.65	86.31	16.40	20.21
DCPO	41.15	91.19	13.33	14.30	DCPO	49.58	91.78	11.04	10.41	DCPO	61.98	78.16	16.38	14.68
+ CalibSFT	41.25	95.07	13.81	17.73	+ CalibSFT	50.21	93.07	12.87	14.52	+ CalibSFT	62.40	85.16	13.29	12.20
C3RL	43.96	79.01	36.25	42.69	C3RL	52.81	84.29	27.21	33.72	C3RL	63.22	67.90	27.80	30.45
+ CalibSFT	43.12	86.74	21.86	25.49	+ CalibSFT	53.65	91.31	15.69	18.30	+ CalibSFT	63.32	80.88	15.65	15.18

Table 7: Per-dataset results on the revised eight-dataset out-of-domain suite. Each question has four sampled responses at temperature 0.6. WebQuestions contains only examples with a single gold answer.

(a) HotpotQA					(b) TriviaQA					(c) DROP				
Method	P@1	AUROC	Brier	ECE	Method	P@1	AUROC	Brier	ECE	Method	P@1	AUROC	Brier	ECE
Base	64.68	57.68	32.82	32.16	Base	55.24	75.80	35.68	36.93	Base	80.36	59.13	18.22	17.03
RLVR	64.58	54.16	33.10	32.17	RLVR	54.97	76.01	40.59	41.40	RLVR	83.53	59.32	15.34	13.31
CalibSFT	65.70	54.18	29.38	20.57	CalibSFT	55.21	65.05	28.44	20.98	CalibSFT	79.92	59.38	31.60	28.93
RLCR	65.80	59.10	25.97	19.77	RLCR	55.24	81.66	23.84	23.30	RLCR	81.44	63.34	14.72	5.43
+ CalibSFT	66.27	59.23	24.25	10.85	+ CalibSFT	55.25	81.53	18.89	10.71	+ CalibSFT	82.32	74.44	14.99	10.75
ReDoubt	64.83	60.64	23.52	12.11	ReDoubt	55.69	82.96	18.67	13.58	ReDoubt	83.63	63.27	14.76	10.80
+ CalibSFT	64.68	63.08	21.70	1.37	+ CalibSFT	56.14	83.92	19.27	15.58	+ CalibSFT	82.53	75.90	15.60	17.36
CoCA	64.90	56.43	22.77	6.02	CoCA	54.77	77.62	21.63	12.97	CoCA	82.47	61.58	15.03	11.56
+ CalibSFT	66.10	61.57	22.92	11.39	+ CalibSFT	54.89	81.85	20.98	13.16	+ CalibSFT	82.50	73.91	17.95	21.99
DCPO	66.15	57.88	25.33	18.77	DCPO	54.55	79.86	25.93	23.91	DCPO	83.45	59.34	13.25	2.30
+ CalibSFT	65.97	60.70	24.43	14.92	+ CalibSFT	55.12	82.56	17.59	8.25	+ CalibSFT	80.42	75.92	16.54	13.64
C3RL	65.38	52.47	33.39	33.12	C3RL	53.76	73.01	35.61	38.53	C3RL	82.65	54.56	16.37	16.07
+ CalibSFT	66.50	58.94	28.45	19.78	+ CalibSFT	54.11	71.36	22.75	12.50	+ CalibSFT	77.44	64.78	23.65	22.03
(d) MuSiQue					(e) LiveBenchReasoning					(f) NQOpen				
Method	P@1	AUROC	Brier	ECE	Method	P@1	AUROC	Brier	ECE	Method	P@1	AUROC	Brier	ECE
Base	45.84	62.90	41.91	42.33	Base	52.88	61.81	40.53	40.59	Base	21.12	66.31	66.04	70.59
RLVR	47.18	58.12	46.19	46.75	RLVR	29.25	60.32	61.75	64.59	RLVR	24.13	67.07	68.92	71.55
CalibSFT	47.19	62.25	26.51	13.95	CalibSFT	53.00	67.67	25.40	17.74	CalibSFT	21.64	59.59	27.33	29.16
RLCR	47.77	66.21	25.68	15.35	RLCR	31.75	68.00	28.49	29.09	RLCR	23.32	70.48	43.84	52.43
+ CalibSFT	49.97	67.87	23.22	6.13	+ CalibSFT	42.25	68.41	21.94	7.75	+ CalibSFT	23.80	72.24	26.44	30.89
ReDoubt	46.90	66.37	24.07	11.21	ReDoubt	25.75	68.44	23.51	24.86	ReDoubt	24.67	70.28	32.15	39.19
+ CalibSFT	48.52	70.04	22.42	5.52	+ CalibSFT	42.25	80.49	19.95	12.38	+ CalibSFT	24.10	70.60	26.80	31.99
CoCA	46.69	67.03	23.69	10.18	CoCA	32.12	66.75	25.79	23.64	CoCA	22.76	66.59	35.28	43.26
+ CalibSFT	48.76	66.88	23.56	7.72	+ CalibSFT	41.50	68.03	23.24	7.96	+ CalibSFT	23.72	70.77	24.86	28.69
DCPO	47.34	65.48	28.28	20.53	DCPO	31.13	67.14	32.10	33.12	DCPO	23.69	69.80	46.27	53.69
+ CalibSFT	48.96	70.49	22.75	8.55	+ CalibSFT	37.75	71.77	21.12	8.83	+ CalibSFT	23.58	73.13	22.65	25.97
C3RL	45.55	63.39	39.88	40.04	C3RL	33.00	57.18	53.55	55.46	C3RL	22.43	64.77	63.63	68.49
+ CalibSFT	47.47	66.39	24.46	11.87	+ CalibSFT	44.62	53.33	28.11	22.78	+ CalibSFT	22.08	61.18	22.25	17.54
(g) PopQA					(h) WebQuestions					(i) Overall (OOD)				
Method	P@1	AUROC	Brier	ECE	Method	P@1	AUROC	Brier	ECE	Method	P@1	AUROC	Brier	ECE
Base	20.85	74.94	57.89	64.10	Base	21.88	67.10	68.16	71.80	Base	45.36	65.71	45.15	46.94
RLVR	21.37	72.78	68.14	71.96	RLVR	26.02	65.76	68.29	70.41	RLVR	43.88	64.19	50.29	51.52
CalibSFT	21.00	63.88	24.80	26.68	CalibSFT	23.59	59.02	32.52	35.25	CalibSFT	45.91	61.38	28.25	24.16
RLCR	21.29	76.88	37.71	47.90	RLCR	25.07	69.40	48.86	56.11	RLCR	43.96	69.38	31.14	31.17
+ CalibSFT	21.16	76.55	24.76	32.22	+ CalibSFT	24.63	70.20	31.41	36.33	+ CalibSFT	45.71	71.31	23.24	18.20
ReDoubt	21.37	76.83	26.82	36.23	ReDoubt	26.84	70.22	37.08	43.81	ReDoubt	43.71	69.88	25.07	23.98
+ CalibSFT	21.25	77.87	24.54	32.32	+ CalibSFT	24.91	69.57	29.11	34.59	+ CalibSFT	45.55	73.93	22.42	18.89
CoCA	21.06	73.92	30.84	39.88	CoCA	24.65	64.40	38.35	45.46	CoCA	43.68	66.79	26.67	24.12
+ CalibSFT	20.88	76.99	24.12	31.76	+ CalibSFT	26.06	69.71	25.93	28.47	+ CalibSFT	45.55	71.21	22.95	18.89
DCPO	20.99	77.02	41.10	49.96	DCPO	24.59	68.48	50.52	57.04	DCPO	43.99	68.13	32.85	32.41
+ CalibSFT	21.18	76.52	21.29	28.02	+ CalibSFT	25.61	70.87	27.45	30.32	+ CalibSFT	44.82	72.75	21.73	17.31
C3RL	20.81	71.98	54.60	61.92	C3RL	23.94	60.19	67.27	70.35	C3RL	43.44	62.19	45.54	48.00
+ CalibSFT	20.66	66.98	16.56	12.95	+ CalibSFT	23.94	61.29	24.87	19.55	+ CalibSFT	44.60	63.03	23.89	17.38

- **CalibSFT.** Follow our official implementation, the target combines group success rate with response correctness, $c_i^* = 0.5q_x + 0.5z_i$.

All variants initialize from Qwen3-8B and use the same supervision objective. Reasoning and answer tokens receive supervision on correct responses, while confidence tokens receive supervision on all responses. We train for 80 steps (40 for SaySelf) with a batch size of 256 and a learning rate of 2×10^{-6} . We report the proportion of training targets in each of 10 equal-width confidence intervals for all five methods in Table 8.

E.3 TRAINING STABILITY

[TODO:] Show training curves and report the mean and standard deviation of evaluation metrics across training seeds.

E.4 ADDITIONAL METRICS

[TODO:] Report the additional calibration and selective-prediction metrics, including MacroCE, adaptive ECE, PCE, and AURC/EAURC, together with their definitions and aggregation protocols. Include ECE sensitivity to 10, 15, and 20 bins, and plots comparing calibration metrics with AUROC and Pass@1.

E.5 GENERALIZATION TO ADDITIONAL MODELS

[TODO:] Evaluate CalibSFT and representative downstream RL objectives with OLMo and Gemma, using the same ID/OOD protocol as the main experiment.

E.6 CONFIDENCE VISUALIZATION

[TODO:] Show reliability diagrams and confidence distributions for models with and without CalibSFT initialization.

E.7 CONFIDENCE DISTRIBUTION ANALYSIS

[TODO:] Report Top-1 and Top-3 confidence mass and confidence entropy, with per-dataset results and ID/OOD averages.

E.8 TRAINING EFFICIENCY

[TODO:] Compare CalibSFT and confidence-aware RL using wall-clock time, GPU-hours, optimization steps, and generated rollout/token counts.

E.9 QUALITATIVE CASE STUDIES

[TODO:] Include representative high-confidence errors, low-confidence correct answers, confidence corrections preserved after RL, and failure cases.